Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

Sr.No. 6739

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:202 Course Section:1	Seat No: 2022600006

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Name of the candidate. Vinayak Bhatia

Candidate's UID number: 2 0 2 2 6 0 0 0 0 6

Examination class room number: 2 0 2

Invigilator's signature with date: 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE-AIML

DAY AND DATE: Wednesday 8/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

RE/2025/03

6813

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

Q3] a] → Recursive Gradient Equation $\frac{\partial L}{\partial h_{t-k}}$ for a Recurrent Neural Network.

• For a RNN, we have hidden state given by

$$h_t = \tanh(w_{ax} x_t + w_{hh} h_{t-1} + b_h)$$

where w_{ax} is weight matrix of input

w_{hh} is weight matrix of previous hidden state

& output is given by

$$\boxed{y = w_o \times h_t + b_o}$$

w_o ⇒ weight matrix for output gate

Now ~~to~~ ~~a~~ loss function for RNN would be

$$\boxed{\lambda = \frac{\partial L}{\partial h_{t-1}}}$$

$$\frac{\partial L}{\partial h_{t-1}} = \frac{\partial L}{\partial h} \times \frac{\partial h}{\partial h_{t-1}}$$

$$\therefore = \frac{\partial l}{\partial h} \times \frac{\partial h}{\partial h_{t-1}} \Rightarrow (\tanh(w_{ax} x_t + w_{hh} h_{t-1} + b_h))$$

here we have tanh in the multiplication term which gives values between -1 to 1.

→ Over period of time, this effectively reduces gradient of loss function to a very small number leading to vanishing gradient problem

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→ This is where LSTM comes into place

for an LSTM we have cell state c_t which keeps track of what information to remember & what to forget

$$\boxed{c_t = f_t \cdot h_{t-1} + i_t \odot \vec{c}_t}$$

NOW

for the gradient of loss function here we

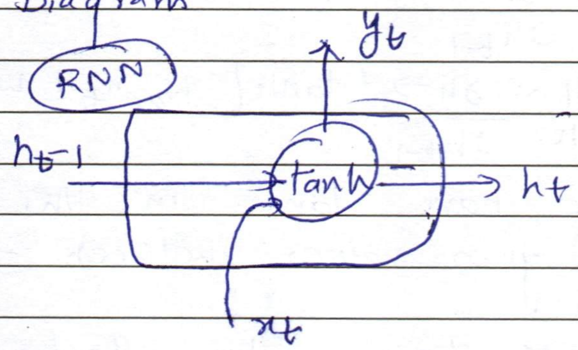
$$\frac{\partial L}{\partial c_{t-1}} = \frac{\partial L}{\partial c_t} \times \frac{\partial c_t}{\partial c_{t-1}}$$

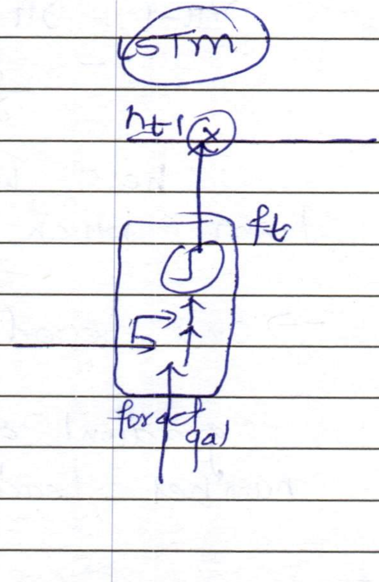
$$= f_t$$

∴ here we are multiplying gradient with forget gate. Now when forget gate $= 1$, it remembers the ~~hidden~~ previous input & prevents the vanishing gradient problem

However when forget gate $= 0$, it forgets the previous input.

Diagram





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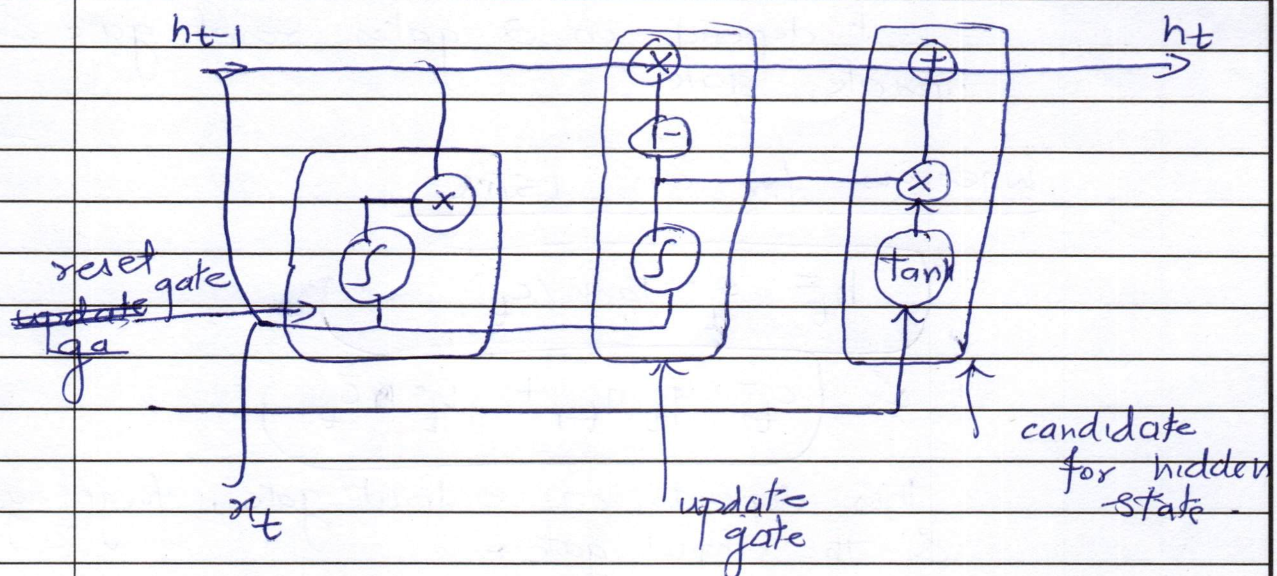
Q3] b]

Gating Equations of a Gated Recurrent unit



For a GRU

We have a reset gate

equation of reset gate $\Rightarrow \boxed{\sigma(W_h h_{t-1} + W_x x_t + b)}$

as its combination of hidden state & input

then we have a update gate

$\boxed{\sigma(W_h h_{t-1} + W_x x_t + b_u)}$

Then we have, $\boxed{\tilde{h}_t = \tanh(W_x x_t + r_t \odot h_{t-1} + b_c)}$

candidate for hidden state

Now finally the hidden state.

$$\boxed{h_t = (1-z_t)\cdot h_{t-1} + z_t \cdot \tilde{h}_t}$$

output of update gate

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∴ As seen from the equations

$$\boxed{h_t = (1-z)h_{t-1} + z\bar{h}_t}$$

$$\bar{h}_t = \tanh(w_x x_t + r_t \odot h_{t-1} + b)$$

$\Downarrow$

It depends on 2 gates reset gate & update gate.

Whereas for an LSTM

$$\boxed{h_t = o_t \cdot \tanh(c_t)}$$

& $\boxed{c_t = f_t h_{t-1} + i_t \cdot \bar{c}_t}$

∴ This depends on output gate, forget gate & the input gate.

Hence the GRU's tend to have fewer parameters than LSTMs.

This can practically be useful in improving generalization in scenarios where there is resource constraint or desire for a light weight architecture

eg stock Market Prediction, real-time sentiment analysis for a new startup. etc.

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Q2]
b] Linear undercomplete Autoencoder learns sparsed representation of the input data. In the forward pass we

$$z_t = \sigma(W_x x_t + b)$$ → encodes the input to dimensions of the compressed hidden state with fewer neurons.

It is called the bottleneck layer which has fewer neurons.

→ This particularly helps to compress the data or forces model to capture rich & important features from the data.

→ This is particularly useful for dimensionality reduction.

→ This latent ~~sp~~ subspace sometimes tends to replicate the PCA architecture which effectively reduces higher dimensional data to a lower dimensional data.

The output again for a undercomplete autoencoder is

$$o_t = \sigma(W_2 z_t + b_2)$$

& the MSE or the loss error is

$$L = \| x - \hat{x} \|^2$$ ⇒ which is the exact same as that of a PCA network.

→ overcomplete autoencoders have higher dimensional hidden layer than the input layer. If allowed to remain in this state, it effectively copies the input as it as it has sufficient neurons to learn the exact input.

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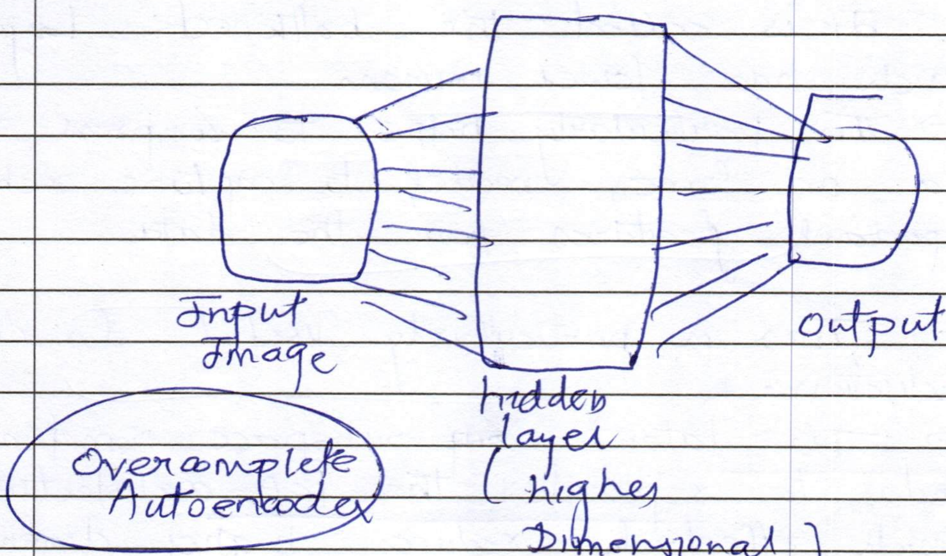
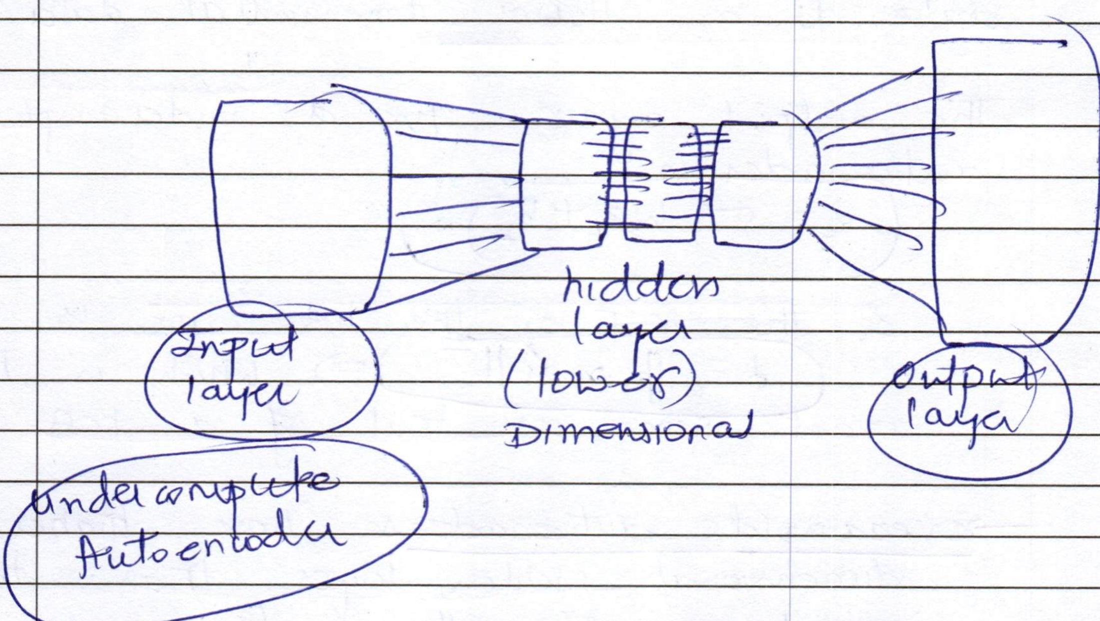
→ Without specific regularization constraint ie: L_1 or L_2 regularization penalty term added to the loss function, it cant be used effectively

→ Overcomplete autoencoders are particularly useful when used with a sparsity constraint & used in cases like denoising of an noisy image etc.





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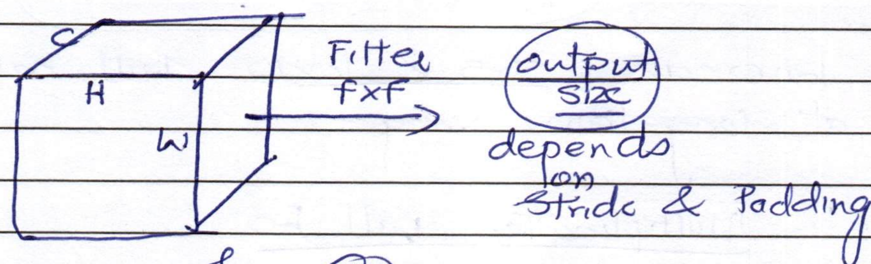
Q2]

a] In a Standard 2-D convolution

we have an input image of size $H \times W \times C$

with a filter given of size $F \times F$.



for eg

38×38 image with Stride = 2
& Padding = 2

$$\text{Output} = \frac{38 - F + 2(2)}{2} + 1$$

let $F = 10 \times 10$

$$O = \frac{38 - 10 + 4}{2} + 1$$

$O = 17$

= 17x17 sized image

⇒ 17x17x10

we simply slide over the filter on the input image & do average pooling or max-pooling

$$\text{total slides} \Rightarrow \boxed{\left(\frac{H}{F}\right) \times \left(\frac{W}{F}\right) \times C}$$

No of channels

Whereas for depthwise seperable convolutions

we have filters of different size to capture features at different levels.

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Suppose we have filter of size

1x1, 3x3, 5x5 & 10x10

∴ 1x1 & 3x3 ⇒ will captur information at minute scale in-depth feature at Shorter range

Whereas 5x5 & 10x10 will captur featur at long-term range.

∴ Multiplication will be

$$\boxed{\left(\frac{H}{F}\right) \times \left(\frac{W}{F}\right) \times K} \Big/ C \Rightarrow \text{depth}$$

K ⇓ K-times we are doing

Why its Important

↳ capturing information at different ranges is very crucial especially in complex scenarios as some features are embedded deep into the image which might not be easier to detect with filter of bigger size.

→ Also the parameter size plays a crucial role in real-life scenarios

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Q1]
a]

$$\boxed{f(w) = \frac{1}{2}\|xw - y\|^2}$$

Gradient descent update rule

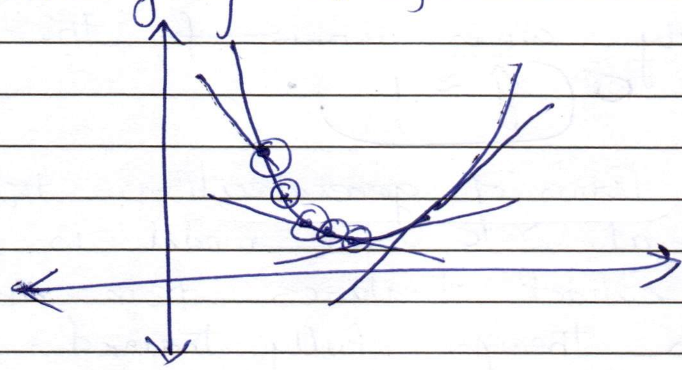
$$\boxed{f'(w) = \|xw - y\| \times w}$$

$\cancel{\|xw-y\|}$ (

$\boxed{\eta < \frac{2}{\lambda_{max}}}$ $\lambda_{max} \Rightarrow$ largest eigenvalue of $x^T x$

, as $\frac{2}{\lambda_{max}}$ would be the minimum threshold

$\Rightarrow$ This is the condition for convergence for a gradient descent to prevent it from going to far



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Q1]
b]

Squared error of a predictor

$$\boxed{\cancel{E = \|x-\hat{x}\|^2}}$$

$$\Downarrow$$

$$\boxed{E = \|y-\hat{y}\|^2}$$

$\downarrow$ decomposing it into bias-variance decomposition.

When models capacity is too high

→ it effectively tries to copy the input without learning good representation of the input data. In this case the error comes close to 0 as model simply copies ~~input~~ of the input as it is. $\boxed{\hat{y} \approx y}$.

→ In terms of generalization, this plays a crucial role as model is not able to extract features from an input data even though fully trained.

→ This usually happens when hidden layer dimension is significantly high & there is no sparsity constraint to it.

→ For this, we usually calculate the bias-variance decomposition of the error to understand if the model is trained well or is overfitting.

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